

Statistical Learning Final Presentation

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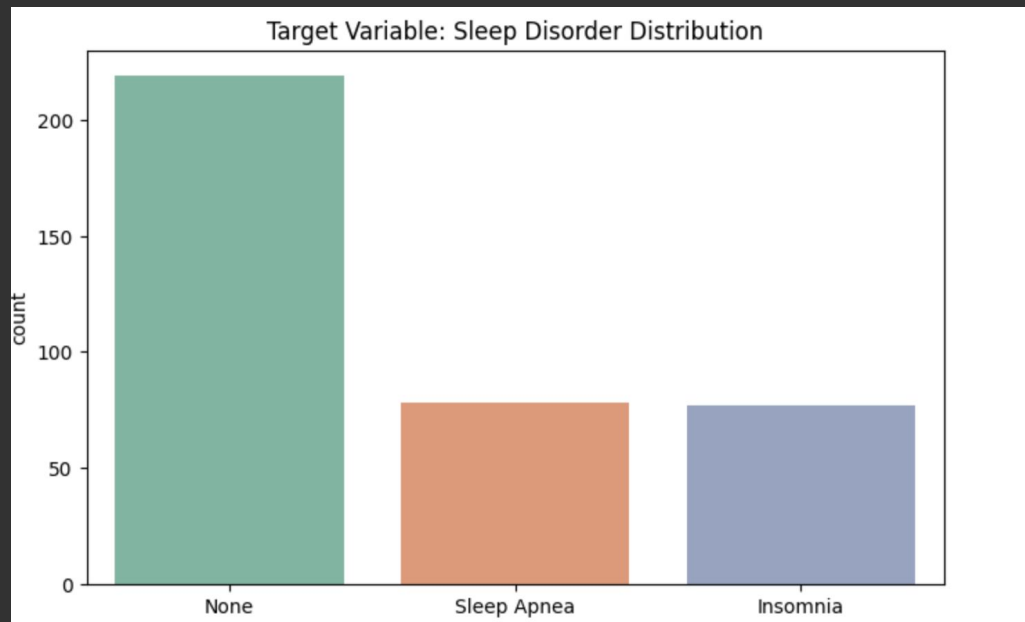
Dataset Overview

Source: Sleep Health and Lifestyle Dataset (Kaggle)

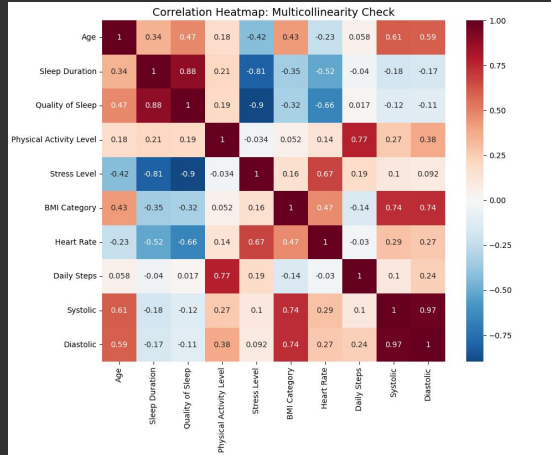
Observations: 374 total with 13 variables

Visually: "None" is our majority class

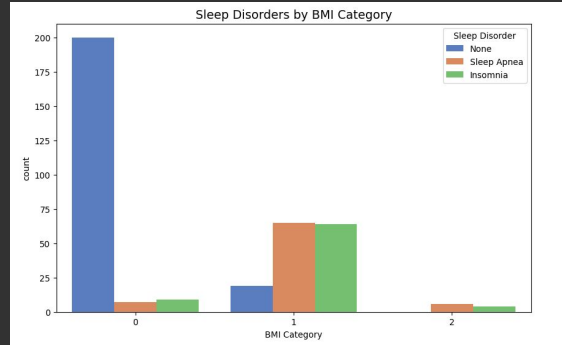
Key features: Age, Stress Level (1-10), BMI Category, and Blood Pressure



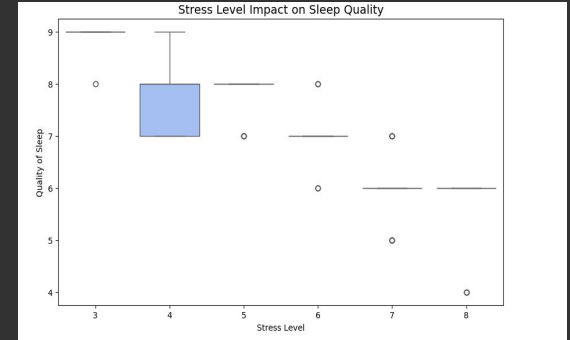
EDA Highlights



We identified significant multicollinearity between Stress and Sleep Quality via our heatmap. To ensure our Inference was robust, we relied on Random Forest, which handles these redundant features better than a standard linear model.



With our BMI analysis, we noticed that weight category isn't just a variable, it's a gatekeeper. We see a nearly perfect absence of Sleep Apnea in the normal weight group, while the overweight group sees a massive surge. This separation played into our Random Forest achieving such high accuracy. The 'decision boundaries' between these health profiles are distinct in the real world.



There is a clear, measurable penalty that stress level exerts on sleep quality. Even at moderate stress, patients with disorders never reach the peak sleep quality of healthy individuals. This 'ceiling' on sleep quality is what our models are picking up on to distinguish between a healthy person and a patient with a disorder.

Methods Overview

K-Nearest Neighbors

- **Non-Parametric Baseline:** Similar lifestyles equals similar disorder groups
- **Optimization:** Determined $K=11$ as the optimal neighbor count.
- **Requirement:** Standard Scaling to prevent high-magnitude features (Steps) from drowning out BMI.

Logistic Regression:

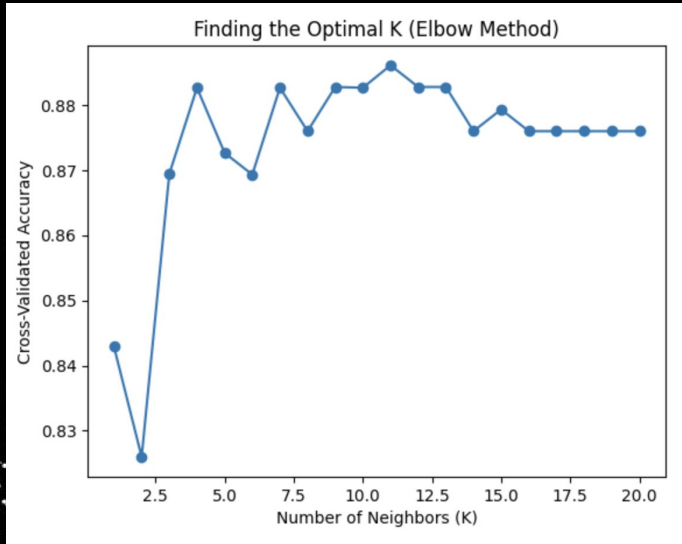
- **Inference Power:** Used to establish a statistical baseline
- **Odds Ratio:** For all of the health risk factors
- **Application:** Quantified the "Stress Penalty" on sleep and isolated BMI as a primary driver for Sleep Apnea.

Random Forest

- **Predictive Lead:** To handle non-linear relationships.
- **Tuning Results:** Optimized with 50 estimators and a depth of 7. Proved robust against class imbalance.
- **Validation:** Paired T-Test confirmed random forest's superiority.

Prediction Results

K Nearest Neighbors Results



Hyperparameters/Tuning:

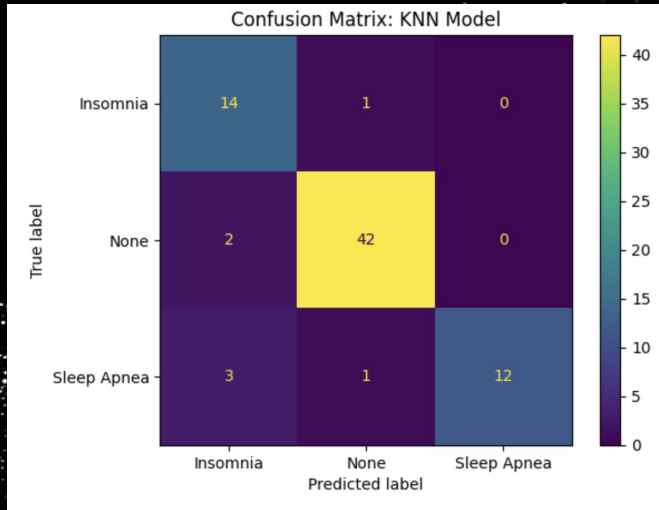
- Found the Optimal K based on all features

Feature Selection:

- All 21 Features

K Nearest Neighbors Results

	precision	recall	f1-score	support
Insomnia	0.74	0.93	0.82	15
None	0.95	0.95	0.95	44
Sleep Apnea	1.00	0.75	0.86	16
accuracy			0.91	75
macro avg	0.90	0.88	0.88	75
weighted avg	0.92	0.91	0.91	75



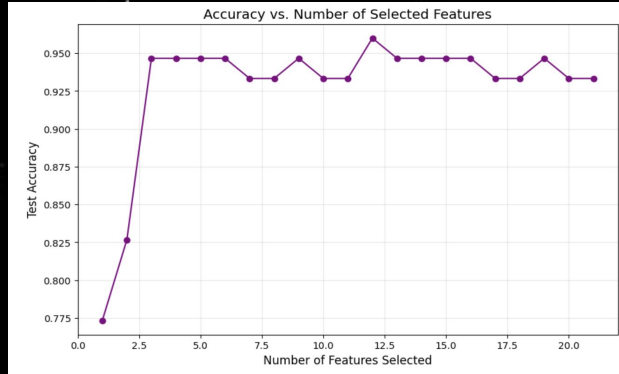
Performance Metrics:

- Accuracy: 0.91
- F1-Scores:
 - None - 0.95
 - Insomnia - 0.82
 - Sleep Apnea - 0.86

Key Findings:

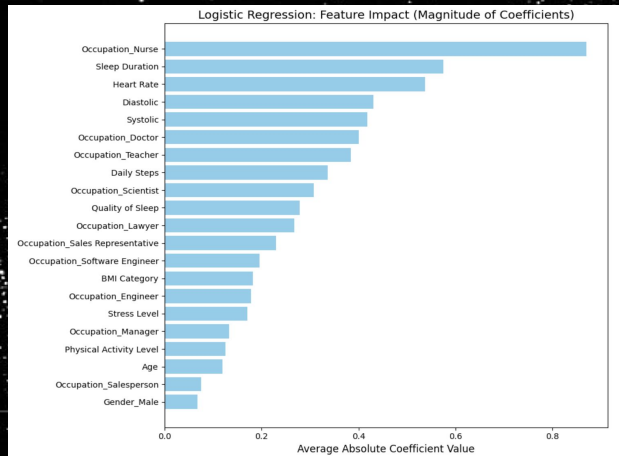
- Correctly identified 42 out of 44 healthy ("None") cases.
- Correctly identified 14 of 15 "Insomnia" cases.
- Struggled with "Sleep Apnea," identifying only 12 of 16 cases correctly.

Logistic Regression Results



Hyperparameters/Tuning:

- Found the highest accuracy based on the selected # of features



Feature Selection:

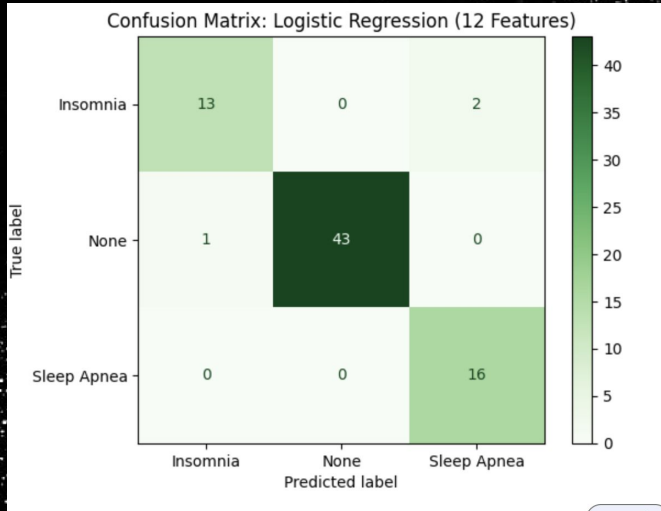
- Sleep Duration, Heart Rate, Daily Steps, Systolic, Diastolic, and seven occupation indicator variables (Doctor, Engineer, Lawyer, Nurse, Sales Representative, Scientist, Teacher)

Logistic Regression Results

	precision	recall	f1-score	support
Insomnia	0.93	0.87	0.90	15
None	1.00	0.98	0.99	44
Sleep Apnea	0.89	1.00	0.94	16
accuracy			0.96	75
macro avg	0.94	0.95	0.94	75
weighted avg	0.96	0.96	0.96	75

Performance Metrics:

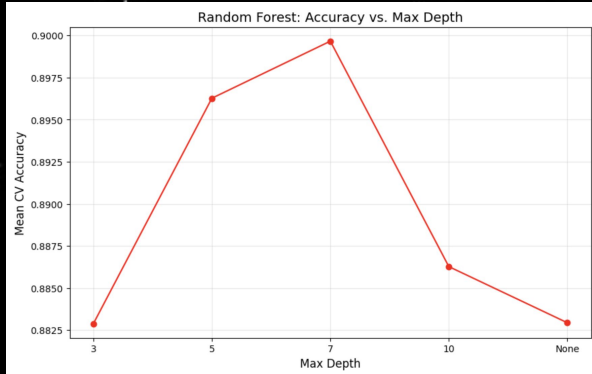
- Accuracy: 0.96
- F1-Scores:
 - None - 0.99
 - Insomnia - 0.90
 - Sleep Apnea - 0.94



Key Findings:

- Perfect performance with "Sleep Apnea," correctly identifying all 16 cases.
- Correctly identified 43 of 44 "None" cases, with only 1 false positive.
- Correctly predicted 14 of 15 "Insomnia" cases.

Random Forest Results

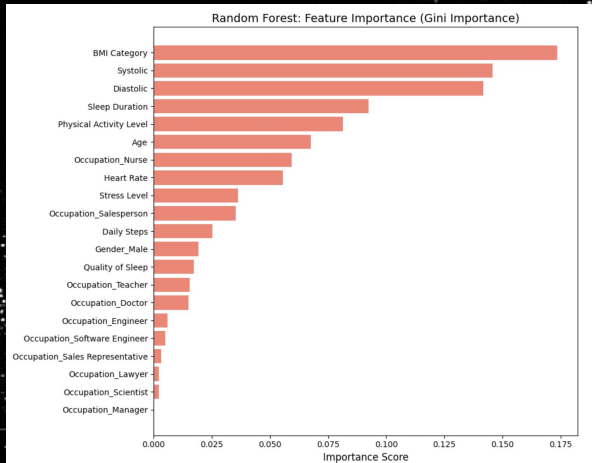


Hyperparameters/Tuning:

- Found the Highest Accuracy based on the Max Depth

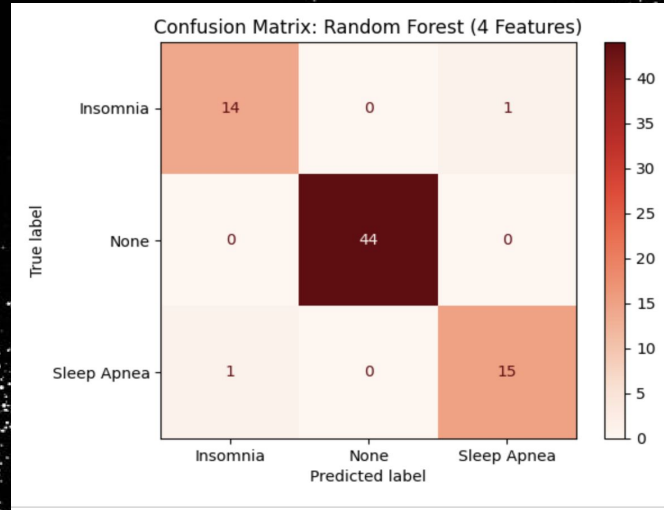
Feature Set:

- Sleep Duration, BMI Category, Systolic, and Diastolic



Random Forest Results

Random Forests Final Classification Report:				
	precision	recall	f1-score	support
Insomnia	0.93	0.93	0.93	15
None	1.00	1.00	1.00	44
Sleep Apnea	0.94	0.94	0.94	16
accuracy			0.97	75
macro avg	0.96	0.96	0.96	75
weighted avg	0.97	0.97	0.97	75



Performance Metrics:

- Accuracy: 0.973
- F1-Scores:
 - None - 1.00
 - Insomnia - 0.93
 - Sleep Apnea - 0.94

Key Findings:

- Correctly identified all 44 "None" cases, resulting in zero false negatives for healthy patients.
- Achieved the highest accuracy using the fewest features (4 vs. 12).
- Successfully identified 14 of 15 "Insomnia" cases and 15 of 16 "Sleep Apnea" cases.

Initial Best Model and Practical Implications

Practical Implications:

- False Negatives (Missing a Diagnosis) - Both Logistic Regression and Random Forest resulted in 0 of these
- False Positives (Incorrect Diagnosis) - Random Forest was the only one to not have any false positives regarding healthy patients.

Best Model:

- Based on our results, we would choose the Random Forest model. It had the highest accuracy and was able to predict all None cases perfectly. While it did have errors, it was for predicting a disorder for another disorder.

Inference Results

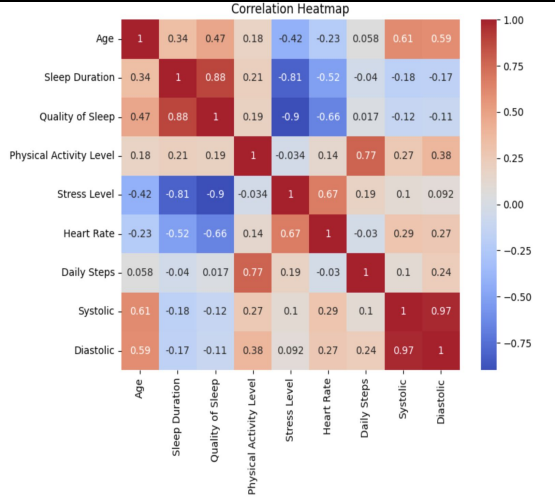
Inference Model Results

Model Used:

- Multinomial Logistic Regression
- Used to identify which health and lifestyle factors are most associated with sleep disorders

Heatmap:

- Heatmap shows relationships between numeric variables
- Sleep Duration and Quality of Sleep had a strong positive correlation
- Stress Level had a negative relationship with sleep quality and duration
- Systolic and Diastolic Blood Pressure were highly correlated



Statistical Metrics:

- Pseudo R² of 0.635 suggests the model had a fairly strong fit
- LLR p-value is extremely small, meaning the model was statistically significant overall
- Model successfully converged, so the results are reliable for interpretation

Key Findings:

- Bottom table shows coefficients, odds ratios, and p-values
- Heart Rate was the most significant predictor across categories
- Systolic and Diastolic Blood Pressure were strongly associated with sleep disorders
- Daily Steps had a measurable effect
- Age was statistically significant across multiple categories

	coef_0	coef_1	odds_0	odds_1	pval_0	pval_1
Heart Rate	0.160408	0.633421	1.173989	1.884045e+00	0.204124	0.000024
Diastolic	0.542114	1.325351	1.719639	3.763507e+00	0.085014	0.000052
const	-4.874642	-78.430948	0.007638	8.667069e-35	0.733634	0.000064
Systolic	-0.565286	-0.848825	0.568198	4.279173e-01	0.027134	0.000857
Daily Steps	0.000665	0.001731	1.000665	1.001733e+00	0.187705	0.002199

Systemic Method Comparison

Comparing Initial Testing to Cross-Validation

PREDICTION METHOD MODELS COMPARISON

	Method	Features	Test Accuracy	Test F1 (Weighted)	Test AUC-ROC
0	KNN	All (21)	0.9067	0.9076	0.9719
1	Logistic Regression	12	0.9600	0.9600	0.9920
2	Random Forest	4	0.9733	0.9733	0.9978

Initial Testing resulted with
Random Forest having the
highest accuracy

SYSTEMATIC METHOD COMPARISON

	Method	Optimal Config	CV Accuracy (Mean)	CV Accuracy (Std)	CV F1-Score (Weighted)	CV AUC-ROC (Weighted)
0	KNN	All Features (21)	0.8861	0.0316	0.8855	0.9042
1	Logistic Regression	12 Features	0.8929	0.0363	0.8920	0.9002
2	Random Forest	4 Features	0.8863	0.0399	0.8855	0.9051

Once Cross-Validation was performed though we can see that Logistic Regression ended up with the highest Accuracy. KNN had the lowest Standard Deviation making it the most stable.

Method Comparison Table

Metric	KNN	Logistic Regression	Random Forest
Hyperparameters/Tuning	K Neighbors	Regularization (RFE)	Trees, Depth, Feature Engineering
Test Accuracy	0.906	0.960	0.973
Test AUC-ROC	0.972	0.992	0.998
Cross-Validation Accuracy	0.886	0.893	0.886
CV AUC-ROC	0.904	0.900	0.905
False Positives	2 cases	1 case	0 cases
False Negatives	4 cases	0 cases	0 cases

Discussion & Recommendations

Discussion

- Sleep disorders can be predicted effectively using health and lifestyle variables
- Cardiovascular and lifestyle-related features were consistently important across models
- Model performance shows a trade-off between accuracy and interpretability

Recommendations

- Logistic Regression is the best overall model. It's easily interpretable and transparent.
- Explore larger datasets and additional health-related features in future analysis

Interpretability vs. Accuracy

Interpretability:

- Logistic Regression is the best for interpretability in this scenario. For example, we can easily explain how much a one-unit increase in Stress Level affects the likelihood of Insomnia.

Accuracy:

- For accuracy, we saw that Logistic Regression also ended up being the highest, but when done across multiple samples.

Conclusion

- Statistical learning models were effective in predicting sleep disorders
- Random Forest achieved the highest predictive performance
- Logistic Regression provided strong accuracy with better interpretability
- Heart Rate, Blood Pressure, Age, and Daily Steps were key factors associated with sleep disorders
- These models can support early detection and better healthcare decision-making